

Numerical comparison of hybrid stochastic simulation algorithms for reaction kinetics

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Abstract

Biochemical reaction networks with species present in low copy numbers exhibit stochastic effects that a deterministic rate-equation description cannot capture, yet an exact stochastic simulation of the same network can become prohibitively slow once other species reach high abundance. Hybrid algorithms address this by partitioning a network into a discrete part, resolved exactly by the stochastic simulation algorithm (SSA), and a continuous part, advanced by a cheaper approximation such as the chemical Langevin equation (CLE). Duncan et al. proposed such a hybrid framework, based on a smooth, molecule-number-dependent blending weight, as an alternative to the propensity-threshold scheme of Alfonsi et al., but validated it only against a pure SSA or a pure continuous simulation, never directly against another hybrid method under a common setup. This project work provides that comparison: the Alfonsi algorithm and the two variants of the Duncan framework, Duncan-SSA and Duncan-Thinning, are implemented in a common framework and benchmarked against the SSA and the CLE on two test systems, a biochemical Lotka–Volterra model and the bacteriophage T7 replication model, with respect to the accuracy of expectation values and probability distributions, mean runtime, and sensitivity to the respective control parameter. Both hybrid algorithms reproduce the SSA result closely on both test systems, at markedly lower computation time than the SSA, in regimes where the CLE itself deviates substantially. Relative to Alfonsi, the Duncan framework is markedly less sensitive to the choice of its control parameter but this robustness comes at the cost of runtime, since Duncan spends a larger share of the simulation in the stochastic regime and is consequently the slower of the two algorithms. For the test system considered here, the theoretical advantage claimed for the Thinning variant, namely that it permits a larger continuous step size than Duncan-SSA, could not be confirmed: both variants reproduce the SSA distribution up to the same step size.

Introduction

Single-cell measurements have shown that many intracellular processes are inherently noisy: transcription and translation proceed in random bursts, so that genetically identical cells in the same environment can differ substantially in their molecular content [1, 2]. Where a regulatory species is present in only tens to a few hundred copies, this randomness need not average out but can decide an outcome. HIV-1 infected cells with identical proviral genomes split into high and near-zero expression states through stochastic Tat fluctuations amplified by positive feedback [3]. Bacterial persistence is a second case: stochastic switching into a slow-growing state lets a subpopulation survive antibiotics without carrying any resistance mutation [4]. All these examples couldn't be described correctly using deterministic reaction kinetics which shows the importance of choosing the right modelling framework depending on the number scales involved in the network.

In systems biology a biochemical reaction network is treated on one of three scales. On the *macroscopic* scale, where every species is abundant, the network is modelled by deterministic reaction-rate equations, a system of ordinary differential equations for continuous concentrations. On the *microscopic* scale, where some species are present in only tens or hundreds of molecules, this description fails: the copy numbers are discrete, the random timing of individual reactions produces fluctuations that are no longer small relative to the mean, and a trajectory can reach states the deterministic model never predicts, such as the extinction of a species. The dynamics is then governed by the chemical master equation (CME) for the probability of each discrete state. Between the two lies the *mesoscopic* scale, where the network is again advanced by differential equations for the macroscopic averages but with a superimposed noise term that reproduces the intrinsic fluctuations, the best-known example being the chemical Langevin equation (CLE) [5].

The microscopic description is the most fundamental of the three, and the one the others approximate. Its master equation can rarely be solved in closed form, so trajectories are instead generated by Monte Carlo simulation. Gillespie's stochastic simulation algorithm (SSA) [6] does this exactly, but it advances the state one reaction event at a time, so its cost is set by the number of events a trajectory contains – and that number grows quickly as soon as some reactions fire very frequently or some species reach high copy numbers. Cheaper approximations – τ -leaping [7] and, in the continuous limit, the chemical Langevin equation [5] – stop resolving individual events; the master equation, the SSA and both approximations are set out in the Methodology. They are accurate, however, only where every reaction fires many times per step and molecule numbers stay well away from zero, that is, not for the rare species that made a stochastic treatment necessary in the first place.

Hybrid methods address this by exploiting the fact that a realistic network usually contains both regimes at once, and that a species may move between them as the simulation proceeds. Instead of committing the whole network to a single description, they let every reaction be treated in the regime it actually belongs to: reactions involving rare species keep their exact, event-by-event resolution, the remainder is advanced by the cheap approximation, and the two descriptions are coupled into one trajectory. This work compares two such schemes. In the adaptive method of Alfonsi et al. [8] each reaction is assigned to its regime by a propensity threshold, and the continuous part follows the deterministic rate equation. In the blending scheme of Duncan et al. [9] each reaction can belong to three different regimes – stochastic, continuous or hybrid – and the continuous part follows the CLE. A reaction in the hybrid regime contributes to both parts at once, with a share that shifts gradually as the molecule numbers of its species change. This blending scheme is studied in two of its variants, Duncan-SSA and Duncan-Thinning. What neither scheme has so far been measured against, however, is the other one. Both were benchmarked by their authors against the exact SSA and against a purely continuous simulation, but no study puts them side by side – neither in accuracy nor in computational cost – on the same models with the same settings.

That is the gap this project work addresses. Both schemes are run on two benchmark networks from the literature – the biochemical Lotka–Volterra system used by Duncan et al. [9] and the bacteriophage T7 replication model of Srivastava et al. [10] – with the exact SSA as ground truth. Four questions guide the comparison: how closely each scheme reproduces the SSA expectation values and copy-number distributions, how much computing time it needs to do so, how strongly the outcome depends on the control parameter

that governs the partition, and – for the two Duncan variants – how large a continuous step each of them can still sustain. The report is organized as follows: the Methodology section recalls the stochastic-kinetics background, defines the two algorithms, and introduces the test models and numerical experiments; the Results section reports the accuracy–cost comparison for each model, the effect of the control parameter, and the comparison of Duncan-SSA with Duncan-Thinning; the Discussion interprets the findings, states the limitations, and outlines possible extensions.

Methodology

Theoretical background: stochastic reaction kinetics

We consider a well-stirred reaction system of N chemical species in a fixed volume at constant temperature. Its state at time t is the vector of molecule numbers $\mathbf{X}(t) \in \mathbb{N}_0^N$. The dynamics are driven by R reactions; reaction j is specified by its state-change vector $\boldsymbol{\nu}_j \in \mathbb{Z}^N$ and its propensity function $\alpha_j(\mathbf{x})$, where $\alpha_j(\mathbf{x}) dt$ is the probability that reaction j fires exactly once in the interval $[t, t + dt]$, given $\mathbf{X}(t) = \mathbf{x}$. When j fires, the state moves to $\mathbf{x} + \boldsymbol{\nu}_j$. At small molecule numbers a description by smooth, continuous concentrations is no longer adequate: the random fluctuations are no longer small relative to the molecule numbers themselves and a trajectory can reach configurations that a deterministic model never would – such as the extinction of a species. A stochastic description is then necessary.

The complete information about the system is contained in the probability $P(\mathbf{x}, t)$ of residing in state $\mathbf{x}$ at time t . Its time evolution obeys the chemical master equation

$$\frac{\partial P(\mathbf{x}, t)}{\partial t} = \sum_{j=1}^R \left[\alpha_j(\mathbf{x} - \boldsymbol{\nu}_j) P(\mathbf{x} - \boldsymbol{\nu}_j, t) - \alpha_j(\mathbf{x}) P(\mathbf{x}, t) \right]. \quad (1)$$

This is a linear system of differential equations, but its state space is usually unbounded and a closed-form solution exists only for a few special cases such as purely monomolecular networks, so in practice one has to resort to numerical methods.

Gillespie’s Stochastic Simulation Algorithm (SSA) [6] generates statistically exact realizations of this master equation. Starting from $\mathbf{X}(t) = \mathbf{x}$ with the total propensity $\alpha_0(\mathbf{x}) = \sum_{j=1}^R \alpha_j(\mathbf{x})$, it determines the state after the next reaction event in three steps:

1. Draw the waiting time until the next event, $\tau \sim \text{Exp}(\alpha_0(\mathbf{x}))$.
2. Draw the index j of the firing reaction with probability $\alpha_j(\mathbf{x})/\alpha_0(\mathbf{x})$.
3. Update the state according to $\mathbf{X}(t + \tau) = \mathbf{x} + \boldsymbol{\nu}_j$.

The algorithm is exact, but its cost grows with the number of simulated reaction events. For networks with fast reactions or high molecule numbers the simulation therefore becomes very expensive, which motivates the development of approximate and hybrid methods.

A first approximation is τ -leaping. Over a step of length τ one no longer resolves the individual events but only counts how often each reaction fires. If τ is short enough that the propensities stay essentially constant over the step, the number of firings of reaction j is Poisson-distributed with mean $\alpha_j(\mathbf{x}) \tau$, so that

$$\mathbf{X}(t + \tau) \approx \mathbf{x} + \sum_{j=1}^R \boldsymbol{\nu}_j \mathcal{P}_j(\alpha_j(\mathbf{x}) \tau), \quad (2)$$

with independent Poisson variables $\mathcal{P}_j$ [7].

If the step in addition contains many firings of every reaction, $\alpha_j(\mathbf{x}) \tau \gg 1$, each Poisson variable is well approximated by a normal variable with the same mean and variance, $\mathcal{P}_j(\alpha_j \tau) \approx \alpha_j \tau + \sqrt{\alpha_j \tau} \xi_j$ with

independent standard normal ξ_j . Substituting this, identifying $\sqrt{\tau}\xi_j$ with the increment ΔW_j of a Wiener process, and letting $\tau \rightarrow dt$ turns the update into the chemical Langevin equation, the Itô stochastic differential equation

$$d\mathbf{X}(t) = \sum_{j=1}^R \boldsymbol{\nu}_j \alpha_j(\mathbf{X}(t)) dt + \sum_{j=1}^R \boldsymbol{\nu}_j \sqrt{\alpha_j(\mathbf{X}(t))} dW_j(t), \quad (3)$$

with independent standard Wiener processes $W_j(t)$. The drift term is the deterministic reaction-rate equation, while the diffusion term, with amplitude $\sqrt{\alpha_j}$, reproduces the variance of the discrete firing counts and thus carries the intrinsic noise [5]. The approximation requires a regime in which every reaction fires often and $\mathbf{X}(t)$ stays away from zero; for rare species it breaks down.

Real networks often contain both regimes at once: some species are present in high copy number and can be described continuously, while others are so rare that only an exact discrete simulation reproduces their behaviour correctly. Moreover, a species can switch between the regimes over the course of a trajectory. Hybrid methods therefore partition the reactions into a part that is treated exactly by the SSA and a part that is propagated through a continuous approximation and couple the two descriptions. The algorithms compared in this work differ mainly in how they carry out this partitioning and handle the coupling; they are introduced in the following section.

Hybrid algorithms under comparison

All three methods studied here split the reactions into a discretely and a continuously treated part and they differ in three respects:

- (i) The type of partition: a sharp assignment of each reaction to exactly one regime (Alfonsi) versus a smooth transition with weights $\beta_j \in [0, 1]$ (Duncan);
- (ii) The model of the continuous part: a deterministic rate equation (Alfonsi) versus the chemical Langevin equation (Duncan);
- (iii) The generation of the inter-event times: comparing the integrated total propensity against an $\text{Exp}(1)$ threshold (Alfonsi), drawn from $\text{Exp}(\lambda'_0)$ using the weighted total propensity (Duncan-SSA), or from a thinning procedure with a constant upper rate (Duncan-Thinning).

Alfonsi algorithm

The algorithm of Alfonsi et al. [8] repeatedly partitions the reactions by a propensity threshold α_{thresh} and advances the deterministic part between two stochastic events. Let S denote the current set of stochastic reactions and D the set of deterministic ones. One macro step consists of:

1. Partitioning the reactions according to the current state: reaction j is placed in S if $\alpha_j(\mathbf{x}) < \alpha_{\text{thresh}}$, and in D otherwise, so that reactions may change regime over the course of the simulation.
2. Draw the threshold $\xi \sim \text{Exp}(1)$ for the next stochastic event.
3. Integrate the deterministic subsystem

$$\dot{\mathbf{x}}(t) = \sum_{j \in D} \boldsymbol{\nu}_j \alpha_j(\mathbf{x}(t)) \quad (4)$$

with a stiff solver, carrying along the accumulated stochastic propensity $\Phi(\tau) = \int_0^\tau \alpha_S(\mathbf{x}(s)) ds$ with $\alpha_S = \sum_{j \in S} \alpha_j$, until $\Phi(\tau) = \xi$.

4. At that time, draw the firing reaction $j \in S$ with probability $\alpha_j(\mathbf{x})/\alpha_S(\mathbf{x})$ and update the state to $\mathbf{x} + \boldsymbol{\nu}_j$.

Blending scheme of Duncan et al.

The blending scheme of Duncan et al. [9] is the common basis of the two remaining methods. Rather than assigning each reaction to a single regime, it lets every reaction contribute to both the discrete and the continuous update, with a weight that depends smoothly on the current molecule numbers: a reaction among abundant species is advanced entirely by the CLE, a reaction that involves a species present in small numbers entirely by SSA, and a reaction in between by both parts in proportion.

The weight is set per species. For each species n a blending interval $[I_1^n, I_2^n]$ is prescribed, and a blending function $\beta(x_n)$ decreases from 1 to 0 as the molecule number x_n grows across this interval. Below I_1^n the species counts as rare and $\beta = 1$; above I_2^n it counts as abundant and $\beta = 0$ and between the thresholds β interpolates linearly. Formally,

$$\beta(x_n; I_1^n, I_2^n) = \begin{cases} 1 & x_n \leq I_1^n, \\ \frac{I_2^n - x_n}{I_2^n - I_1^n} & I_1^n < x_n < I_2^n, \\ 0 & x_n \geq I_2^n. \end{cases} \quad (5)$$

A reaction is treated discretely as soon as any of its participating species ($\nu_{nj} \neq 0$) is rare, so its weight combines the species weights as

$$\beta_j(\mathbf{x}) = 1 - \prod_{n: \nu_{nj} \neq 0} (1 - \beta(x_n; I_1^n, I_2^n)), \quad (6)$$

This weight splits the propensity of each reaction into a discrete and a continuous part:

$$\lambda_j'(\mathbf{x}) = \beta_j(\mathbf{x}) \alpha_j(\lfloor \mathbf{x} \rfloor), \quad \lambda_j''(\mathbf{x}) = (1 - \beta_j(\mathbf{x})) \alpha_j(\mathbf{x}). \quad (7)$$

The discrete parts λ_j' act as the propensities of a pure stochastical step that proposes the next reaction and its time; the continuous parts λ_j'' form the drift and diffusion of one CLE step over the interval up to that time. A purely stochastic reaction ($\beta_j = 1$) contributes only to the stochastical step, with its full propensity α_j ; a purely deterministic reaction ($\beta_j = 0$) contributes only to the CLE step, likewise with its full propensity. A reaction in the intermediate range contributes to both: it is proposed as a discrete jump at the reduced rate $\lambda_j' = \beta_j \alpha_j$ and at the same time enters the CLE step with the complementary weight $\lambda_j'' = (1 - \beta_j) \alpha_j$. Equivalently to Duncan et al. the CLE progression has been implemented by the weak trapezoidal method proposed by Anderson et al. [11].

Both variants use two step sizes with $\Delta t \leq \delta t$: δt in the pure CLE regime and Δt as the upper bound for the continuous step in the transition regime.

Duncan-SSA

The hybrid SSA method (as the name suggests) treats the discrete part by pure SSA dynamic (Algorithm 1). In the transition regime the waiting time τ is drawn from the instantaneous rate $\lambda_0'(\mathbf{x}) = \sum_j \lambda_j'(\mathbf{x})$. This is only admissible as long as λ' stays approximately constant over τ , which keeps Δt small.

Algorithm 1 Duncan-SSA (after `hybrid_duncan_ssa`)

Input: $\mathbf{S}$, c , $\mathbf{x}_0$, $[t_0, t_f]$, $\{[I_1^n, I_2^n]\}$, δt , Δt with $\Delta t \leq \delta t$

```
1:  $t \leftarrow t_0$ ;  $\mathbf{x} \leftarrow \mathbf{x}_0$ 
2: while  $t < t_f$  do
3:    $\beta \leftarrow \beta(\mathbf{x})$ 
4:   if  $\max_j \beta_j = 0$  then ▷ pure CLE regime
5:      $\mathbf{x} \leftarrow \text{CLE}(\mathbf{x}, \delta t)$ ;
6:      $t \leftarrow t + \delta t$ 
7:   else if  $\min_j \beta_j = 1$  then ▷ pure SSA regime
8:      $\lambda' \leftarrow \beta \cdot \alpha(\mathbf{x})$ ;  $\lambda'_0 \leftarrow \sum_j \lambda'_j$ 
9:      $\tau \sim \text{Exp}(\lambda'_0)$ ; draw  $\nu_j$  with  $P(j) = \lambda'_j / \lambda'_0$ 
10:     $\mathbf{x} \leftarrow \mathbf{x} + \nu_j$ ;  $t \leftarrow t + \tau$ 
11:   else ▷ transition regime
12:      $\lambda' \leftarrow \beta \cdot \alpha(\mathbf{x})$ ;  $\lambda'_0 \leftarrow \sum_j \lambda'_j$ 
13:      $\tau \sim \text{Exp}(\lambda'_0)$  draw candidate reaction  $\nu_j$  with  $P(j) = \lambda'_j / \lambda'_0$ 
14:     if  $\tau < \Delta t$  then
15:        $\mathbf{x} \leftarrow \text{CLE}(\mathbf{x}, \tau) + \nu_j$ ;  $t \leftarrow t + \tau$ 
16:     else
17:        $\mathbf{x} \leftarrow \text{CLE}(\mathbf{x}, \Delta t)$ ;  $t \leftarrow t + \Delta t$ 
18:     end if
19:   end if
20: end while
21: Output: trajectory  $(t, \mathbf{x})$ 
```

Duncan-Thinning

The thinning method replaces the cumulated propensity by a constant upper bound $\Lambda_0 = \sum_j \Lambda_j$ with $\Lambda_j \geq \sup_{\mathbf{x}} \lambda'_j(\mathbf{x})$ (Algorithm 2). In the transition regime the proposal time is drawn as $\tau \sim \text{Exp}(\Lambda_0)$, which decouples the step length from propensity constancy: the CLE is advanced over the whole interval $[0, \tau]$ in substeps $\leq \Delta t$, with Δt now limited only by the numerical stability of the CLE, which permits larger continuous steps. Because Λ_0 proposes events too often, a proposal is accepted only with probability $\sum_j \lambda'_j(\mathbf{x}') / \Lambda_0$ evaluated at the state $\mathbf{x}'$ reached by the CLE step, and the firing reaction is then drawn from the λ'_j .

Algorithm 2 Duncan-Thinning (after `hybrid_duncan_thinning`)

Input: additionally, upper bounds $\{\Lambda_j\}$ with $\Lambda_j \geq \sup_{\mathbf{x}} \lambda'_j(\mathbf{x})$

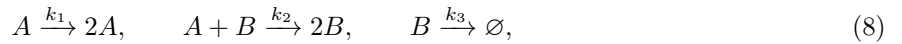
```
1:  $t \leftarrow t_0$ ;  $\mathbf{x} \leftarrow \mathbf{x}_0$ 
2: while  $t < t_f$  do
3:    $\beta \leftarrow \beta(\mathbf{x})$ 
4:   if  $\max_j \beta_j = 0$  then ▷ pure CLE regime
5:      $\mathbf{x} \leftarrow \text{CLE}(\mathbf{x}, \delta t)$ ;
6:      $t \leftarrow t + \delta t$ 
7:   else if  $\min_j \beta_j = 1$  then ▷ pure SSA regime
8:      $\lambda' \leftarrow \beta \cdot \alpha(\mathbf{x})$ ;  $\lambda'_0 \leftarrow \sum_j \lambda'_j$ 
9:      $\tau \sim \text{Exp}(\lambda'_0)$ ; draw  $\nu_j$  with  $P(j) = \lambda'_j / \lambda'_0$ 
10:     $\mathbf{x} \leftarrow \mathbf{x} + \nu_j$ ;  $t \leftarrow t + \tau$ 
11:   else ▷ transition regime: thinning
12:      $\Lambda_0 \leftarrow \sum_j \Lambda_j$ ;  $u_1, u_2 \sim \mathcal{U}(0, 1)$ ;  $\tau \leftarrow -\ln(u_1) / \Lambda_0$ 
13:      $\mathbf{x}' \leftarrow \text{CLE}(\mathbf{x}, \tau)$ 
14:      $\lambda' \leftarrow \beta(\mathbf{x}') \cdot \alpha(\mathbf{x}')$ 
15:     if  $\Lambda_0 u_2 < \sum_j \lambda'_j$  then ▷ accept
16:       draw  $\nu_r$  with  $P(r) = \lambda'_r / \sum_j \lambda'_j$ ;
17:        $\mathbf{x} \leftarrow \mathbf{x}' + \nu_r$ 
18:     else ▷ reject, no jump
19:        $\mathbf{x} \leftarrow \mathbf{x}'$ 
20:     end if
21:      $t \leftarrow t + \tau$ 
22:   end if
23: end while
24: Output: trajectory  $(t, \mathbf{x})$ 
```

Test models

The algorithms are compared on two models taken from the literature, so that the hybrid results can be checked against published data. The Lotka-Volterra model uses the parametrisation of Duncan et al. [9] and the bacteriophage model the respective one from Alfonsi et al. [8].

Lotka–Volterra model

The first model is the biochemical Lotka–Volterra system, two species A (prey equivalent) and B (predator equivalent) with the three reactions



The dimensionless rate constants are $k_1 = 2.0$, $k_2 = 0.002$ and $k_3 = 2.0$, the initial state is $x_A(0) = 50$, $x_B(0) = 60$, and every trajectory is run to $T = 5$, about one oscillation period of the deterministic result. The associated rate equations are the classical Lotka–Volterra model, so the deterministic trajectories can be simulated up to an arbitrary precision for comparison. The species numbers oscillate over roughly four orders of magnitude: large enough near the peaks ($\approx 10^4$) for a continuous description to be advantageous, small enough near the troughs ($\approx 10^1$) for intrinsic noise to matter which can lead to an extinction of species B and a drastic overall change of the system dynamics. On the other side a plain SSA simulation is noticeably expensive in the high molecule regime and especially because α_2 scales with the product $x_A x_B$. The model thus exercises both regimes and the switching between them.

Bacteriophage T7 replication model

The second model describes the intracellular growth of bacteriophage T7 after Srivastava et al. [10]. It has three species – template nucleic acid (tem), genomic nucleic acid (gen) and structural protein (struc) – and the six reactions listed in Table 1. The six reactions are a lumped account of the T7 replication cycle. The injected phage genome is first recruited into a pool of transcriptionally active template (R_1), which then directs the synthesis of both new genomic copies (R_3) and structural protein (R_5); a genomic copy and structural protein are finally packaged, irreversibly, into a progeny virion (R_4), while template and structural protein are removed by degradation (R_2 , R_6). Srivastava et al. introduced the model to show that an infection beginning, as is typical, from a single genome molecule is not captured by the deterministic rate equations: the averaged stochastic kinetics differ from them qualitatively, and at low multiplicity of infection the cell population splits into a productively infected fraction and a persistently low-level infected fraction that can act as a viral reservoir.

Table 1. Reactions of the bacteriophage T7 replication model [10]. State-change vectors ν_j are listed in the species order (tem, gen, struc).

	Reaction	Rate constant	ν_j
R_1	gen $\xrightarrow{c_1}$ tem	$c_1 = 0.025$	$(1, -1, 0)$
R_2	tem $\xrightarrow{c_2} \emptyset$	$c_2 = 0.25$	$(-1, 0, 0)$
R_3	tem $\xrightarrow{c_3}$ tem + gen	$c_3 = 1.0$	$(0, 1, 0)$
R_4	gen + struc $\xrightarrow{c_4}$ virus	$c_4 = 7.5 \times 10^{-6}$	$(0, -1, -1)$
R_5	tem $\xrightarrow{c_5}$ tem + struc	$c_5 = 1000$	$(0, 0, 1)$
R_6	struc $\xrightarrow{c_6} \emptyset$	$c_6 = 1.99$	$(0, 0, -1)$

The initial state is the low infection level $x_{\text{tem}}(0) = 1$, $x_{\text{gen}}(0) = x_{\text{struc}}(0) = 0$. Template and genomic copy numbers then stay of order one for a long time, so their dynamics are dominated by individual reaction events and the infection can die out entirely – an outcome the deterministic rate equations cannot produce. The fast reaction R_5 ($c_5 = 1000$) meanwhile drives the structural protein to several thousand copies. Very small and very large copy numbers coexist in the same trajectory, which makes the model a standard stiff multiscale benchmark for hybrid methods.

Boundary conditions for the continuous update

A CLE step can move a continuous component below zero, which is meaningless for a molecule count, so the continuous update requires a rule at the boundary $x_n = 0$. Two variants are used here: a *reflective* rule, which replaces a negative value by its absolute value, and an *absorbing* rule, which sets it to zero. Only the reproduction of the Lotka–Volterra expectation values of Duncan et al. in the Results section was carried out once with reflective boundary conditions, to match their setup. All other simulations use the absorbing rule, that is, negative values are set to zero.

Numerical experiments and metrics

Reference results at fixed control parameters

The first block of experiments first establishes the literature reference results against which the remaining experiments are measured. For every model and every simulation method an ensemble of 10^3 independent trajectories is generated, starting from the initial states and running to the time horizons given in the previous section. From each ensemble three quantities are estimated: the expectation values $E[x_n](t)$ of every species, the copy-number probability distributions at selected points in time, and the mean runtime per trajectory. For the Lotka–Volterra reaction system consisting of only two species, the two-dimensional distribution has been determined as well. The exact SSA is run on the same ensemble size and serves as the ground truth, and a pure CLE simulation is added as the continuous-only baseline, so that the two hybrid results can be placed between the two limiting descriptions.

Both hybrid schemes have parameters that are model dependent and they had to be fixed before the reference runs could be carried out. This was done by either choosing the original values from the original papers ($\delta t, \Delta t$) or in a set of short pilot simulations preceding the actual experiments: with a reduced ensemble and a shortened horizon. The Alfonsi threshold α_{thresh} and the Duncan blending window – reduced to a single number by setting $I_2^n = I_1^n + 10$ – were scanned over a range of candidate values. Both algorithms approach the exact SSA as their control parameter is increased, at the price of a growing runtime, so the operating point was chosen as the one leaving the lowest possible runtime that still followed the SSA reference over the whole horizon. The values obtained in this way, together with the remaining numerical settings, are collected in Table 2.

The thinning variant requires one further input that the SSA variant does not: the constant upper bounds $\Lambda_j \geq \sup_{\mathbf{x}} \lambda_j'(\mathbf{x})$ from which its proposal times are drawn. These cannot be obtained analytically here, because the weighted propensities are unbounded on the reachable state space – in the Lotka–Volterra model α_2 grows with the product $x_A x_B$, and x_A itself grows without bound once B has gone extinct. The bounds were therefore determined empirically from similar pilot simulations: during SSA and Duncan-SSA runs the largest weighted propensity λ_j' encountered was tracked separately for each reaction over all visited states, and Λ_j was set to that observed maximum + a safety margin (Table 2). The upper bounds matters in both directions. A bound that is too small silently violates the requirement of the thinning construction, so that events are proposed too rarely and the discrete jump statistics are biased; a bound that is far too large leaves the method correct but turns almost every proposal into a rejection, which costs runtime without advancing the trajectory.

Table 2. Numerical settings of the reference runs. The blending window is given in the species order of the respective model; the ensemble size and the horizon apply to every method.

Model	Method	Setting	Value
Lotka–Volterra	all	ensemble size	10^3
	all	horizon T	5
	Alfonsi	α_{thresh}	500
	Duncan	$[I_1^n, I_2^n]$ (A, B)	25/35
	Duncan, CLE	δt	10^{-2}
	Duncan	Δt	10^{-2}
	Duncan-Thinning	Upper boundaries Λ_j (without added margin)	50., 521.59, 50.
Bacteriophage T7	all	ensemble size	10^3
	all	horizon T	200
	Alfonsi	α_{thresh}	60
	Duncan	$[I_1^n, I_2^n]$ (tem, gen, struc)	15 / 25
	Duncan, CLE	δt	10^{-2}
	Duncan	Δt	10^{-3}
	Duncan-Thinning	Upper boundaries Λ_j (without added margin)	6.15, 3.80, 6.0, 0.287, 1000, 30.2

Both the ensemble averages $E[x_n](t)$ and the slices at which the distributions are estimated are taken from a common time-grid; the slices are $t \in \{1, 2, 3, 4, 5\}$ for the Lotka–Volterra model and $t \in \{10, 50, 100, 150, 200\}$ for the bacteriophage T7 model. The SSA, the Alfonsi and both Duncan algorithm are event driven and produce states at irregular, random times, whereas the CLE advance in steps of prescribed length. They therefore usually don't emit states exactly at the desired grid points and the states evaluated here have been defined as the state of the trajectory with the closest possible time point to the corresponding grid. Since the time differences between two stochastic events are usually fairly low this approximation doesn't introduce significant errors.

A second obstacle to a direct comparison is that the continuous methods do not produce molecule numbers in the strict sense at all. The CLE, and with it every part of a Duncan trajectory that is advanced by a CLE step, moves the state along a real-valued path, so an ensemble of such trajectories populates a continuum

rather than the integer lattice on which the SSA lives. Two distinct roundings have to be kept introduced here. The first one is internal to the duncan algorithm: the discrete part of the Duncan scheme evaluates its propensities at the rounded state, $\lambda'_j = \beta_j \alpha_j(\lfloor \mathbf{x} \rfloor)$, so that the proposal of a discrete jump is well defined. The second one is applied only in the evaluation: for all distribution comparisons the recorded states of the CLE and of the Duncan runs are rounded to the nearest non-negative integer, which places the continuous and the discrete methods on the same support. Where copy numbers are large this rounding is irrelevant, since it displaces a sample by at most half a molecule, far less than the width of the distribution. In the bacteriophage T7 model at early times it is not: template and genomic copy numbers are then still single-digit, so the rounding is necessary for the distributions showing clear maxima at the integer states and therefore to agree with the SSA result Fig 6. The expectation values and the runtime measurements use the unrounded states.

The distributions are estimated in two different ways: the phase-space histogram of the Lotka–Volterra model (Fig 1) is a genuine two-dimensional histogram and show where a trajectory spends its time over the whole horizon. The time-resolved distributions, by contrast are kernel density estimates. A Gaussian kernel is centred on each of the 10^3 samples of the slice and the kernels are summed, with the bandwidth chosen automatically by Scott’s rule, $h \propto \hat{\sigma} n^{-1/(d+4)}$ with the sample standard deviation $\hat{\sigma}$, the sample size n and the dimension d . A kernel density estimate was preferred over a fixed binning because the smooth estimate also allows two methods to be overlaid on a common axis and makes the splitting of the distribution into an oscillating branch and a branch with B extinct immediately visible. The price of this smoothing is a systematic distortion that has to be taken into account when the figures are interpreted. A kernel density estimate presupposes an underlying continuous density, whereas the distributions estimated here are genuinely discrete; the estimate therefore spreads probability mass over non-integer values and, close to the boundary, over negative copy numbers, which are not attainable states in reality. Both effects are visible in the bacteriophage T7 distributions at $t = 10$ (Fig 6), where the discrete states $0, 1, 2, \dots$ appear as separate bumps of finite width and a non-vanishing density extends below zero.

Runtime is reported as the average time per trajectory over the whole ensemble which was rerun to each horizon $t = 1, \dots, 6$ separately instead of reading intermediate times off a single long trajectory. Within one model all methods were timed on the same machine, so that the comparison between them is meaningful; however this was not possible between the two models. After the Lotka–Volterra ensembles which were computed locally – due to the high runtimes – the bacteriophage T7 ensembles had to be parallelized over 32 cores at the HPC cluster. The runtimes are therefore comparable inside the models but not between them. All simulations are implemented in Python; the source code is provided in S1 File. Before being applied to the two test models, both hybrid implementations were verified against two analytically solvable systems, a birth–death process and a reversible dimerisation; that validation is documented in S1 Text. Its subsystem is integrated with a stiff BDF solver. Runtimes are measured with the standard `time` module. The average run times were remarkably slower compared to the ones found by Duncan et al. [9] which shows the advantage of using a compiled language as Julia for such computation heavy tasks.

Parameter sensitivity

The control parameters are prescribed only loosely in the original works, so the second block probes how strongly they affect the outcome absent any further prior knowledge. Both algorithms are run on the Lotka–Volterra model over a range of values – $\alpha_{\text{thresh}} \in \{20, 40, 80, 160, 320, 640, 1280\}$ for Alfonsi, and I_1^n over the same set with $I_2^n = 2 I_1^n$ for Duncan. Recorded per parameter are the number of discrete jumps per trajectory, which characterises the share of the simulation spent in the pure stochastic regime, and the expectation values as a function of the horizon $T \in [1, 5]$.

Thinning versus Duncan-SSA

The third block tests whether thinning admits a larger continuous step. The step size Δt is first determined in two ways on the Lotka–Volterra model: (i) from CLE accuracy, by simulating the copy-number distribution at $T = 5$ for a range of Δt and taking the largest value that still shows no significant deviation from the exact SSA distribution under a Mann–Whitney U test; and (ii) from propensity constancy, by

drawing M states from the phase space (weighted by their residence time along sampled trajectories),
 running n short SSA evolutions of length Δt from each, and testing with a TOST procedure whether the
 mean relative propensity change over the step is equivalent to zero within a prescribed margin $\varepsilon = 0.05$; the
 largest Δt for which equivalence still holds is the limit up to which the propensities may be treated as
 constant. The benefit of thinning is then assessed by simulating the distribution of species A at $T = 5$ for
 several step sizes and comparing it with the exact SSA result: with Δt set by the propensity-constancy limit,
 thinning is advantageous precisely if it still reproduces the correct distribution at step sizes beyond that limit.

Both criteria rest on a statistical test; these are stated here in the form in which they are applied.

The Mann–Whitney U test compares two independent samples – here the $n_1 = n_2 = 10^3$ copy numbers of
 species A at $t = 5$, once from the SSA ensemble and once from the method under test. Its null hypothesis is
 that both samples come from the same distribution, so that $P(X > Y) = P(Y > X)$ for a value X drawn
 from the first and a value Y from the second sample. No normal distribution is assumed, since they are
 bimodal for $t = 5$. The test statistic is $U_1 = R_1 - n_1(n_1 + 1)/2$, which counts the pairs (x_i, y_j) with $x_i > y_j$.
 R_1 is the rank sum of the first sample. Under the null hypothesis U has mean $E[U] = n_1 n_2 / 2$ and variance
 $\text{Var}[U] = n_1 n_2 (n_1 + n_2 + 1) / 12$, and for samples of the present size it is approximately normally distributed,
 so that the two-sided p -value follows from

$$z = \frac{U - E[U]}{\sqrt{\text{Var}[U]}} \quad (9)$$

A deviation counts as significant for $p < 0.05$, and the bound is taken as the largest step size whose p -value
 still lies above that level.

The TOST procedure is used for the second criterion because the statement needed there is that the
 propensities do not change significantly over a step. The null hypothesis asserts a change of at least the
 equivalence margin $\varepsilon = 0.05$. The sample consists of the relative propensity changes
 $d = (\alpha_j(t + \Delta t) - \alpha_j(t)) / \alpha_j(t)$ collected from the M sampled states and the n short SSA runs started from
 each of them, with mean $\bar{d}$ and standard error SE. Two one-sided hypotheses are tested against this sample,
 $H_0^- : \mu \leq -\varepsilon$ and $H_0^+ : \mu \geq +\varepsilon$, through the statistics

$$z^- = \frac{\bar{d} + \varepsilon}{\text{SE}}, \quad z^+ = \frac{\bar{d} - \varepsilon}{\text{SE}}, \quad (10)$$

evaluated with the normal approximation. Equivalence is concluded at the level $\alpha = 0.05$ when both
 one-sided tests reject, that is when $\max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$, which is the same as requiring the 90 %
 confidence interval of μ to lie entirely within $[-\varepsilon, \varepsilon]$. Each reaction is tested separately, and a step size is
 accepted only if equivalence holds for all of them.

Results

Lotka–Volterra model

Fig 1 shows the histogram, over 10^3 stochastic trajectories, of the states reached during the simulation duration $T = [0, 5]$ s – which represents one oscillation period of the deterministic solution. As can be seen the histogram broadly reproduces the deterministic trajectory broadly but unlike it the histogram also carries a high density along the axis $B = 0$. This embodies the purely stochastic phenomenon that in a non-negligible fraction of the runs, species B is driven extinct by the stochastic dynamics before T .

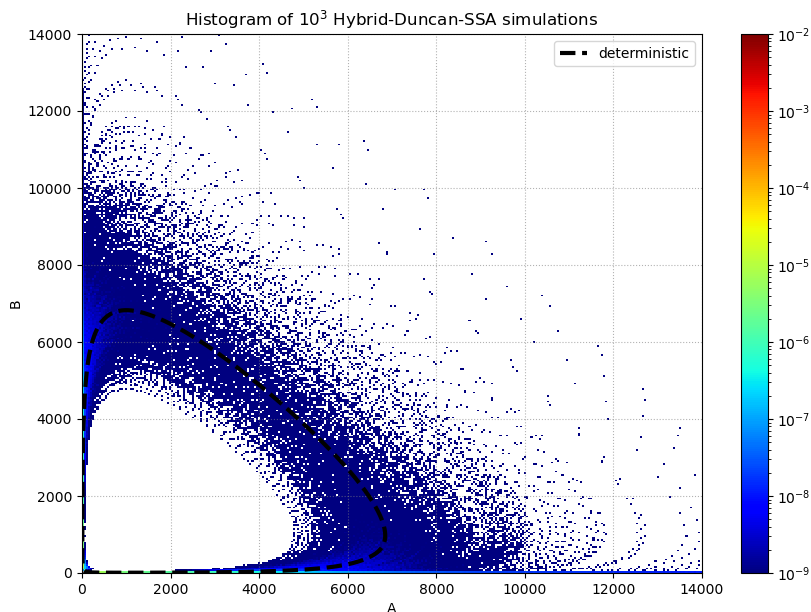
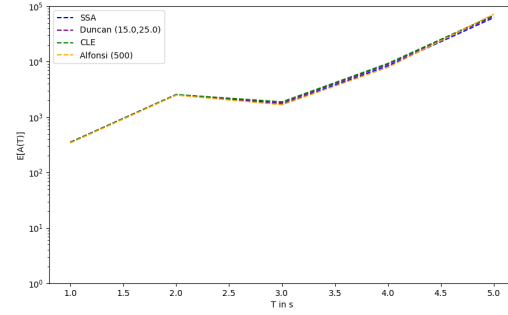
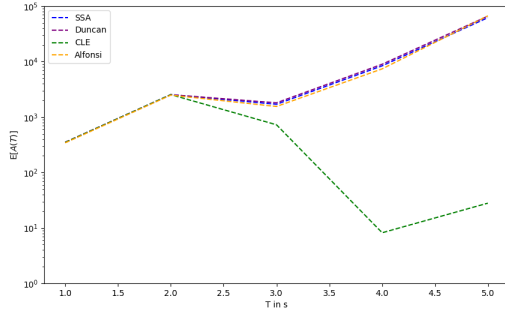


Fig 1. Phase-space histogram of the Lotka–Volterra model. Histogram of 10^3 simulations of the hybrid Duncan-SSA algorithm up to $T = 5.0$ s, starting from $A(0) = 50$, $B(0) = 60$. The dashed line is the solution of the deterministic rate equation.

The result of the simulated expectation value development is shown in Fig 2. It extends the expectation-value comparison of Duncan et al. [9] to include the Alfonsi algorithm. Fig 3 does the same for the average computation time. Over the whole simulated interval, both hybrid methods track the SSA expectation value with only minimal deviation and at markedly lower computation time per simulated duration. Direct comparison for $t = 1$ to 6 indicate no significant difference in the overall time complexity between the two hybrid algorithms.

The CLE simulation, in contrast, depends on how the boundary at zero is handled: with a reflective boundary it departs visibly from the SSA result from $T = 2$ onward and oscillates around the deterministic solution for the rest of the interval, instead of the exponential growth the SSA exhibits (Fig 2a) – reproducing the corresponding result of Duncan et al. [9]. This mechanism is an artifact of the boundary condition itself: a reflective boundary prevents species B from ever truly going extinct, which suppresses exactly the trajectories with exponentially growing A that dominate the SSA mean at later times. Repeating the same simulations with an absorbing boundary (negative molecule-values clipped to zero) removes this artefact: the CLE then agrees closely with the other simulation techniques over the whole simulation duration (Fig 2b).



(a) Reflective boundary.

(b) Absorbing boundary.

Fig 2. Expectation value $E[A]$ for time interval $[1,5]$ under the two boundary conditions. SSA, Alfonsi, Duncan-SSA and the CLE are compared under (a) a reflective and (b) an absorbing boundary at zero.

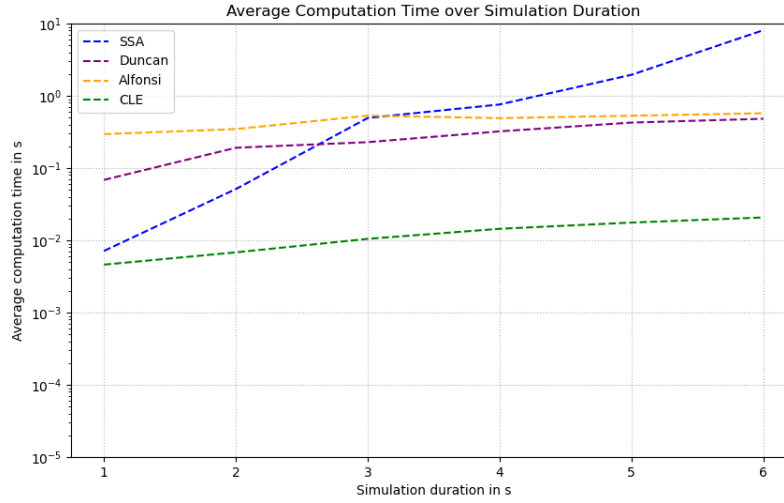


Fig 3. Average computation time, Lotka-Volterra model The average computation time per run (over 10^3 simulations) are shown per simulated time duration, $t = 1$ to 6, again for SSA, Duncan, Alfonsi and CLE.

Besides the expectation values, the full probability distribution have also been simulated over time (Fig 4) as well. The corresponding one-dimensional marginals are given in S1 Fig and S2 Fig). Again from $t = 3$ onward this distribution visibly splits into two branches: trajectories that keep following the oscillation at comparatively small molecule numbers, and trajectories in which B has gone extinct, so that A grows exponentially over $t = 3$ to 5. Also, similar to the results for the expectation values each of the three simulation techniques (CLE, Alfonsi, Duncan) resemble the SSA results with relatively high precision.

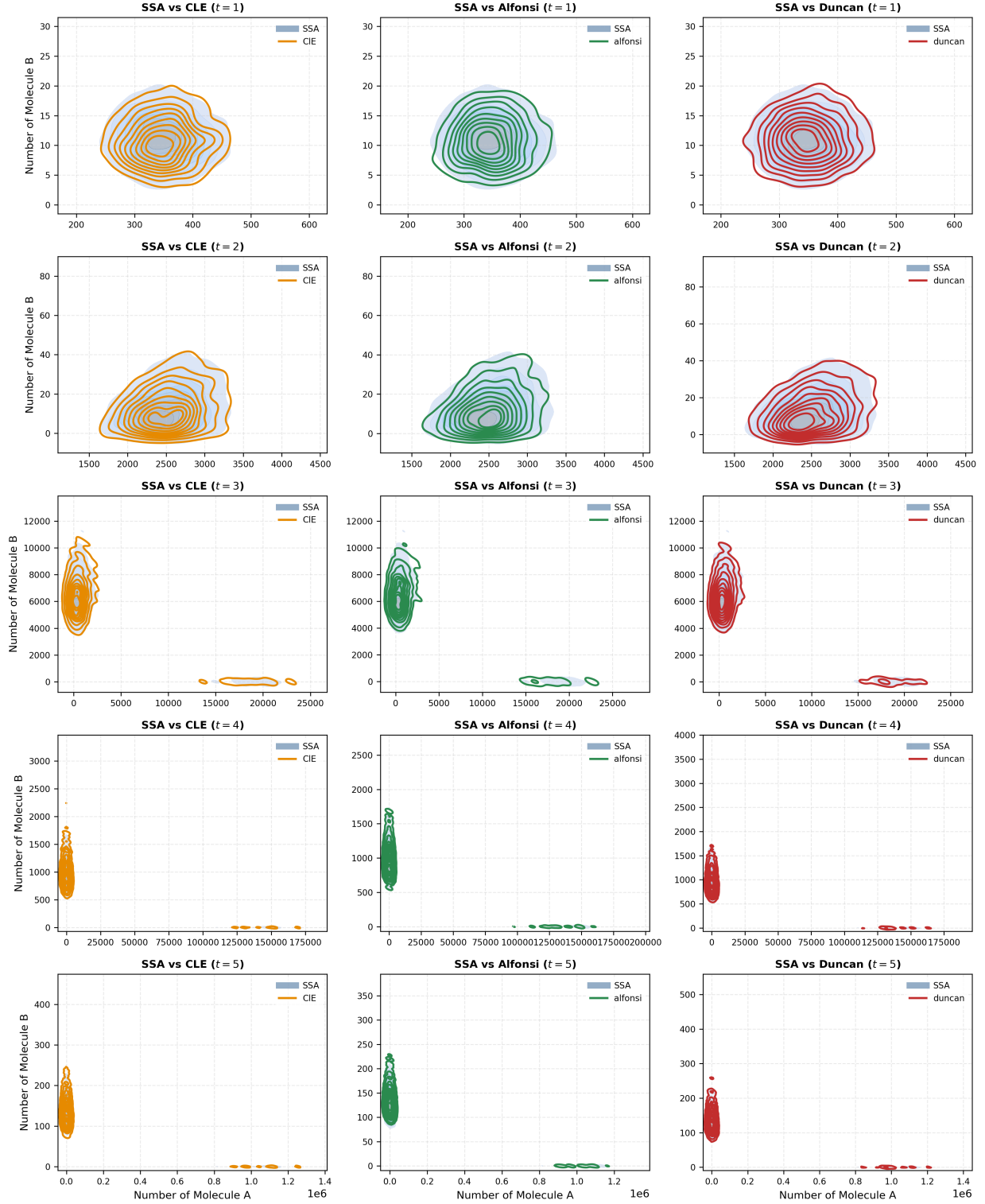


Fig 4. Joint probability distribution of (A, B) over time Simulated distribution at $t \in \{1, 2, 3, 4, 5\}$, comparing SSA, Alfonsi, Duncan-SSA and the CLE using the absorbing boundary condition. From $t = 3$ onward the distribution splits into an oscillating branch at moderate molecule numbers and a branch with B extinct and A growing exponentially.

Bacteriophage T7 replication model

The same three comparisons – expectation value, probability distribution and average computation time – were repeated for the bacteriophage T7 replication model, shown here for the template species (Fig 5, Fig 6). The corresponding expectation values and probability distributions for the genomic and structural species are given in S3 Fig, S4 Fig, S5 Fig and S6 Fig. As for the Lotka–Volterra model, the model shows a clear difference between deterministic result and the expectation of the purely stochastic simulation which justifies the necessity of the latter. For low infection level starting conditions, stochastic simulations lead to random extinction events of the overall virus and therefore to a proportion of trajectories with zero infection which lowers the overall expected level when compared to the deterministic result. Accordingly the probability distributions shows a peak at zero tem-molecules even for high values of t .

The two hybrid algorithms again agree closely with the SSA result while running at markedly lower computation time (Fig 7). Unlike the Lotka–Volterra model, however, the CLE simulation no longer matches the SSA result even under the absorbing boundary condition that removed the discrepancy there: its expectation value instead approaches the deterministic limit at large t . For low t the deviation from the SSA result is even greater than the one from the purely deterministic result. Also the probability distribution estimated from CLE simulation alone clearly lacks the maximum corresponding to extinct trajectories. Another key difference is that two hybrid algorithms also diverge from one another in computation time here, which they did not for Lotka–Volterra: for long simulation times the Alfonsi algorithm is significantly faster, even when compared with the CLE itself.

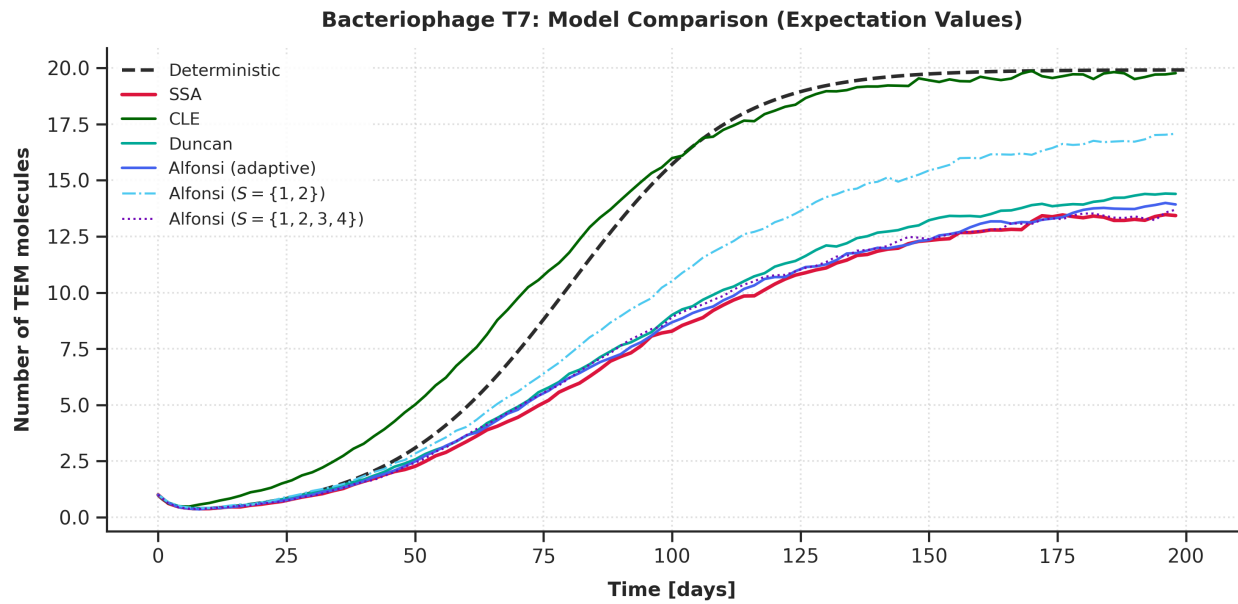


Fig 5. Expectation value of the template species (tem), bacteriophage T7 model. SSA, Alfonsi, Duncan-SSA and the CLE.

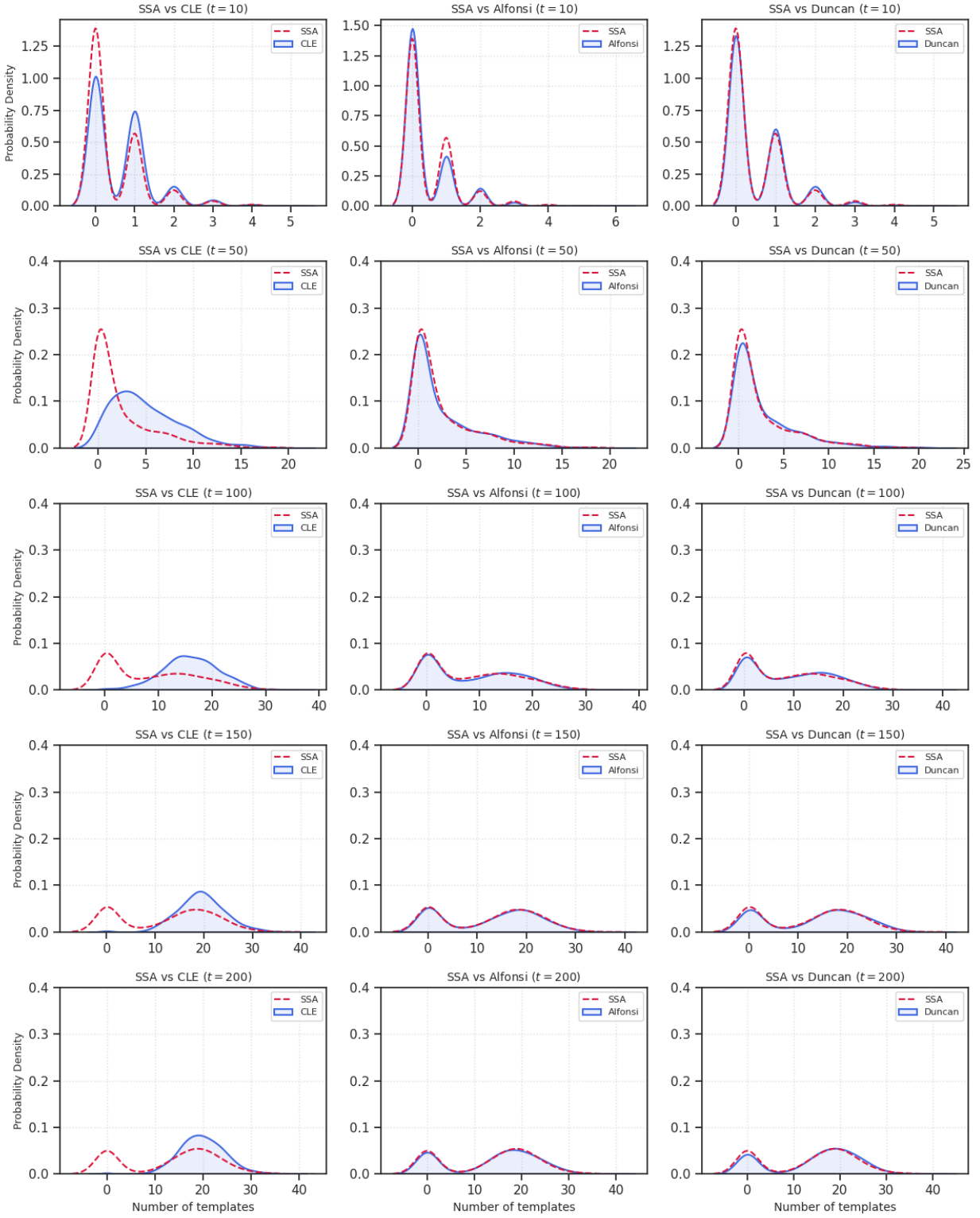


Fig 6. Probability distribution of the template species (tem), bacteriophage T7 model. Alfonsi, Duncan-SSA and the CLE, compared to the SSA result over time $t = [10, 50, 100, 150, 200]$.

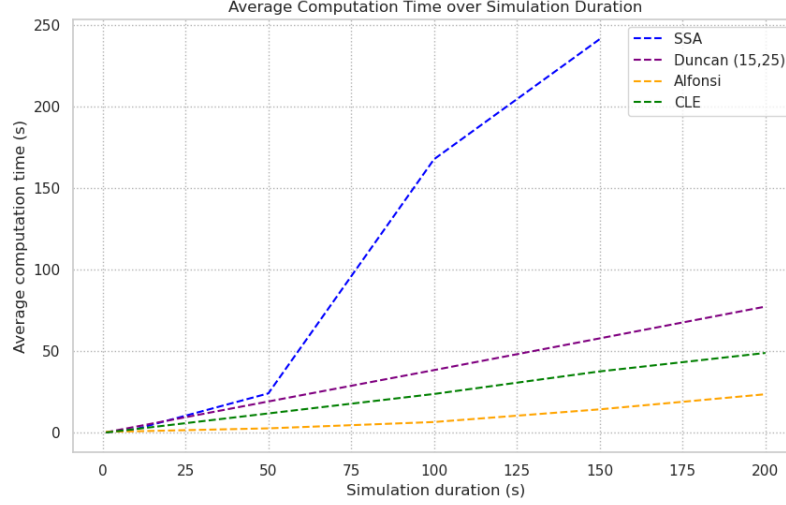
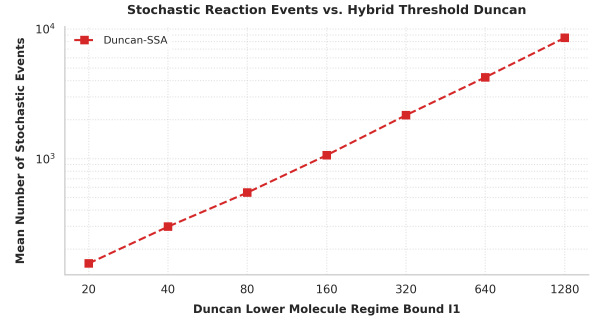
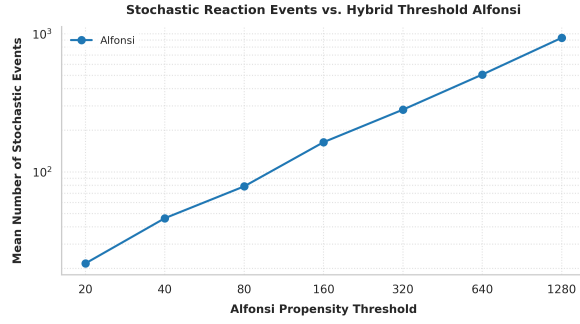


Fig 7. Average computation time per run, bacteriophage T7 model. Comparison between SSA, Alfonsi, Duncan-SSA and the CLE.

Effect of the control parameter: Alfonsi vs. Duncan

Both algorithms depend on a control parameter that governs how reactions are split between the stochastic and the deterministic regime. For Alfonsi this is a single propensity threshold α_{thresh} : a reaction whose propensity falls below the threshold is advanced stochastically, all others deterministically. For Duncan it is the blending window $[I_1^n, I_2^n]$ prescribed per species, which enters the reaction weight β_j through Eq (6). I_1^n, I_2^n are defined as species copy number thresholds that partition species into stochastic, hybrid or deterministic. Here the window is reduced to a single number by fixing $I_2^n = 2 I_1^n$ and varying I_1^n . As a measure of how much of a simulation each algorithm spends in the stochastic regime, the total number of stochastic reaction events over the simulated time was recorded for parameter values $\{20, 40, 80, 160, 320, 640, 1280\}$ (Fig 8). The Alfonsi algorithm produces fewer stochastic events (22 to 935) than Duncan (155 to 8566) – roughly one order of magnitude fewer at every parameter value. Both counts grow with the control parameter in the same way: on the doubly logarithmic axes of Fig 8 the event count follows an almost perfectly straight line for each algorithm, so the two respond to their respective parameter with essentially the same relative sensitivity (approximate slopes $\beta_{\text{alfonsi}} \approx 0.0013$, $\beta_{\text{duncan}} \approx 0.0014$), differing only by the roughly constant offset between them.

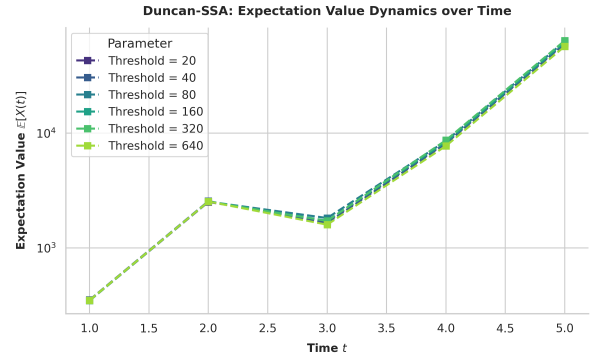
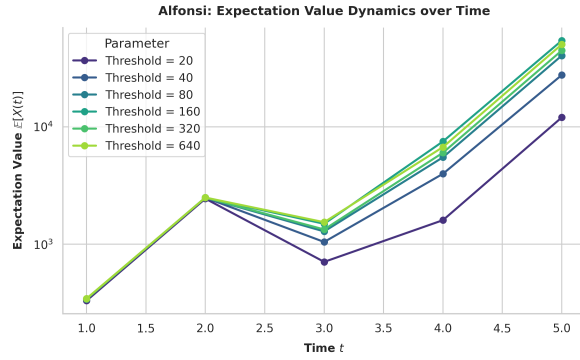
A difference in parameter sensitivity does appear in the expectation values (Fig 9), which show the effect of the parameter choice on the expectation-value trajectory for each algorithm. The Duncan algorithm reproduces the SSA trajectory across its whole parameter range regardless of the exact choice, whereas Alfonsi deviates downward once its threshold is set too small.



(a) Alfonsi.

(b) Duncan.

Fig 8. Number of stochastic reaction events over the parameter range. (a) Alfonsi, propensity threshold α_{thresh} ; (b) Duncan-SSA, lower blending bound I_1^n with $I_2^n = 2 I_1^n$. Duncan mean stochastic events count constant about one order of magnitude above Alfonsi throughout the values. Same Lotka–Volterra model and simulation settings as in the *Lotka–Volterra model* subsection.



(a) Alfonsi.

(b) Duncan.

Fig 9. Effect of the control parameter on the expectation value, Lotka–Volterra model. While Duncan reproduces the SSA result across its whole parameter range; Alfonsi deviates downward once the propensity threshold is set too small. Same model and simulation settings as in the *Lotka–Volterra model* subsection.

Duncan-SSA versus Duncan-Thinning

The upper bound on the CLE step size Δt can be limited by two different effects: the numerical stability of the CLE step itself, or the assumption – common to hybrid algorithms such as Alfonsi and Duncan – that the propensities stay approximately constant over the step. To determine which effect dominates, both bounds were estimated separately.

The bound from CLE stability was obtained by comparing the probability distribution of molecule A between the CLE and the SSA, for step sizes $\Delta t \in \{10^{-2}, 3 \times 10^{-2}, 4 \times 10^{-2}, 5 \times 10^{-2}, 6 \times 10^{-2}\}$. Fig 10 shows the diverging tail of the distribution – where the CLE and the SSA are visually distinguishable – at $t \in \{3, 4, 5\}$; the significance of the deviation, however, was assessed only at $t = 5$, with a Mann–Whitney U test on the full distribution rather than only the plotted tail (Table 3). By this test the deviation becomes significant between $\Delta t = 5 \times 10^{-2}$ and 6×10^{-2} , which is taken as the upper bound set by CLE stability.

Table 3. Mann–Whitney U test comparing the CLE and SSA probability distributions of molecule A (Lotka–Volterra model) at $t = 5$, for different step sizes Δt .

Δt	U -statistic	p -value
10^{-2}	509 135.5	0.4779
3×10^{-2}	513 478.5	0.2949
4×10^{-2}	492 209.5	0.5453
5×10^{-2}	478 516.5	0.0954
6×10^{-2}	462 745.0	0.0039

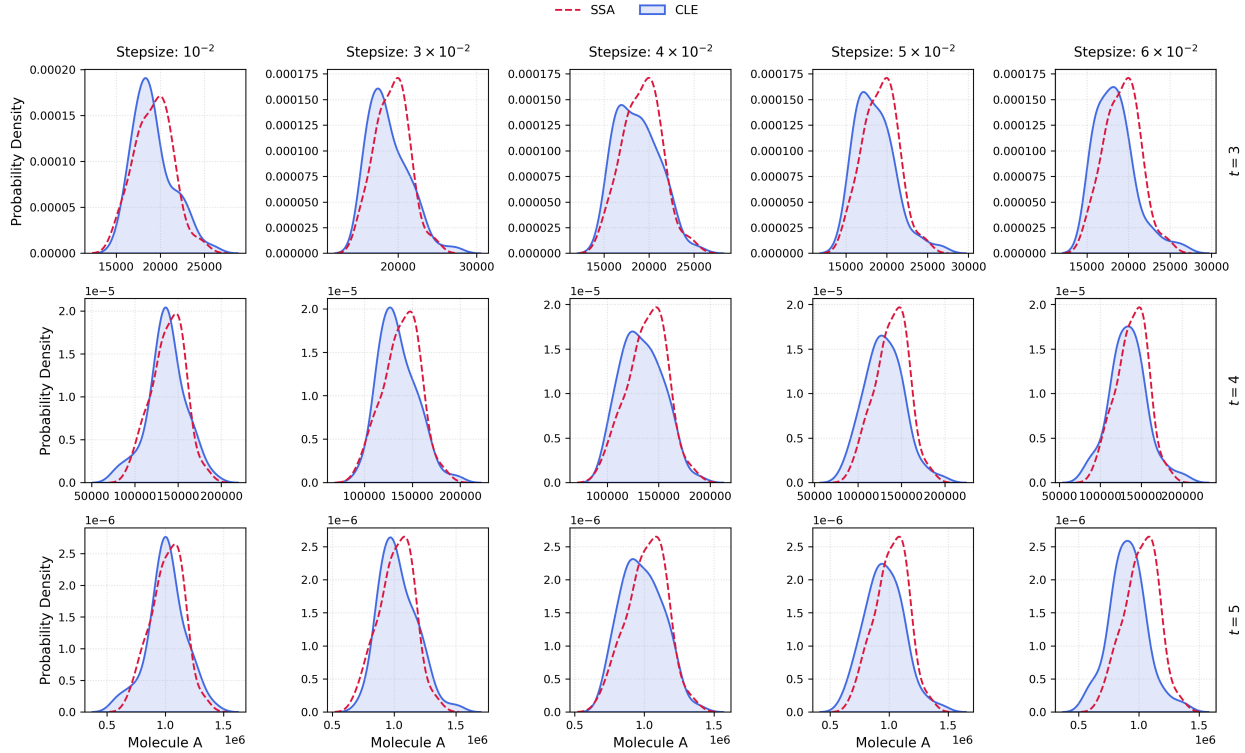


Fig 10. CLE step-size stability, Lotka–Volterra model. Diverging tail of the probability distribution of molecule A at $t \in \{3, 4, 5\}$, CLE versus SSA, for $\Delta t \in \{10^{-2}, 3 \times 10^{-2}, 4 \times 10^{-2}, 5 \times 10^{-2}, 6 \times 10^{-2}\}$.

The bound from propensity constancy was obtained differently. 1000 random states were drawn along the non-diverging trajectories, weighted by their residence time, to obtain a representative sample across phase space. From each state, 1000 SSA simulations were run from 0 up to a candidate step size $\Delta t \in \{10^{-3}, 3.16 \times 10^{-3}, 10^{-2}, 3.16 \times 10^{-2}, 10^{-1}\}$, and the weighted relative propensity difference $\frac{\alpha_t - \alpha_0}{\alpha_0}$ was compared against the equivalence margin $[-0.05, 0.05]$ with a TOST equivalence test. The result, given in S1 Tab, shows a significant change in propensity from $\Delta t = 3.16 \times 10^{-2}$ onward (for two of the three reactions) – a lower step-size bound than the one set by CLE stability above.

Third Duncan-Thinning was checked against Duncan-SSA in the same way as the cle-stability based step-size: for $\Delta t \in \{10^{-4}, 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$, the probability distribution of molecule A (Lotka–Volterra model) obtained with each method was compared against the SSA distribution (Fig 11). Both methods agree with the SSA up to $\Delta t = 10^{-2}$ and deviate visibly beyond it; the same Mann–Whitney U test (Table 4) confirms this quantitatively, and shows no advantage of Duncan-Thinning over Duncan-SSA at any of the tested step sizes.

Table 4. Mann–Whitney U test comparing Duncan-Thinning and Duncan-SSA against the SSA probability distribution of molecule A (Lotka–Volterra model), for different step sizes Δt .

Δt	Method	U -statistic	p -value
10^{-4}	Duncan-Thinning	485 696.5	0.267
	Duncan-SSA	503 430.0	0.790
10^{-3}	Duncan-Thinning	494 314.0	0.659
	Duncan-SSA	510 481.0	0.415
10^{-2}	Duncan-Thinning	563 743.0	3.5×10^{-7}
	Duncan-SSA	548 457.0	1.1×10^{-4}
5×10^{-2}	Duncan-Thinning	200 744.5	6.3×10^{-119}
	Duncan-SSA	200 744.5	6.3×10^{-119}

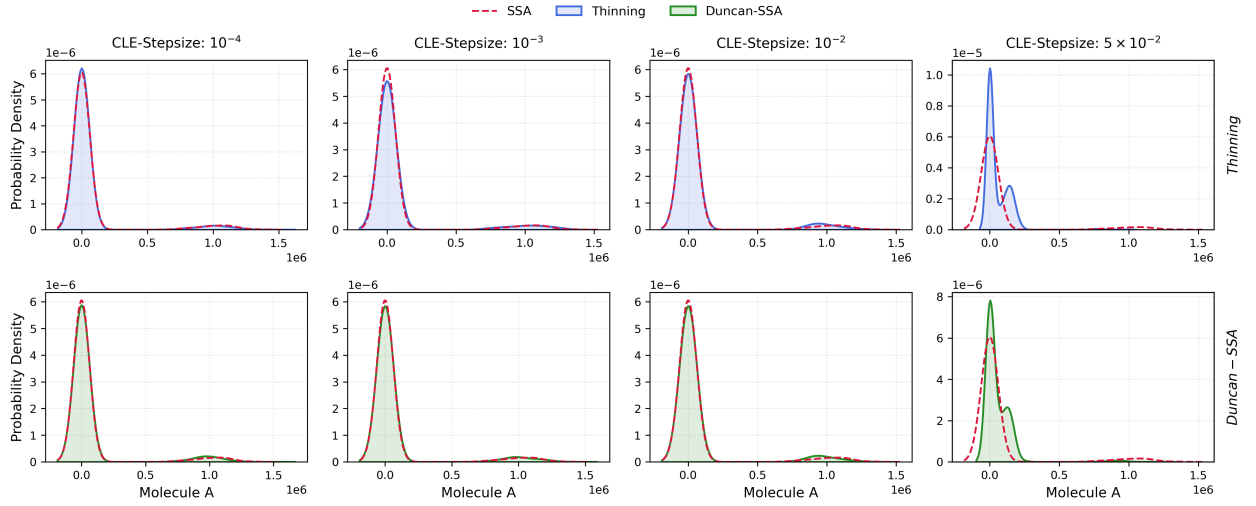


Fig 11. Duncan-Thinning versus Duncan-SSA, Lotka–Volterra model. Probability distribution of molecule A obtained with each method, compared against the SSA distribution, for $\Delta t \in \{10^{-4}, 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$.

Discussion

Both hybrid algorithms reproduce the SSA result on the two test systems considered here with high accuracy while running at lower computation time, and both continue to do so for the bacteriophage model, where the CLE deviates strongly. That being said for the Lotka–Volterra model the CLE deviation shown in the Results is a pure artefact of the boundary condition rather than a shortcoming of the CLE approximation itself: once the boundary at zero is handled by absorption instead of reflection, the CLE already agrees with the SSA. This raises the question whether the Lotka–Volterra system, at the parameters chosen by Duncan et al. [9], is a meaningful test case for a hybrid method at all – the advantage of a hybrid scheme lies precisely in regimes where the continuous approximation breaks down, and here it does not. The bacteriophage T7 model, where the CLE fails to match the SSA even under the absorbing boundary, is in this sense the more informative of the two. Boundary handling remains a central difficulty for CLE simulations in general which is proven once again by the results of this work and motivates the necessity of hybrid modelling framework like the ones analyzed here.

In the limit of high decision parameter values both hybrid algorithms reduce exactly to the SSA even though they do so in different ways. For Duncan, the limit of the parameter results in a setting where every reaction is classified as purely stochastic ($\beta_j = 1$ for all j): the CLE contribution then vanishes and each reaction contributes with its full propensity α_j , which is the exact SSA. For Alfonsi, in the limit no reaction exceeds the propensity threshold, so that the continuous part is empty and the solver only advances the total propensity, held constant over the step, to the next reaction time – exactly the definition of the SSA (see the hybrid step function in `stochastinetics.py`, S1 File). Because this means the hybrid algorithms can reproduce the SSA to arbitrary accuracy for a suitable parameter choice, this accuracy alone is not a good basis for comparing them. What differs between the two is the runtime (for the same accuracy): and as was shown the older Alfonsi algorithm achieves the same result at comparable (Lotka–Volterra) or lower (bacteriophage) computation time.

In another section Duncan et al. argue that their scheme is, relatively insensitive to the choice of its control parameter – a practical advantage, since that parameter cannot be determined in advance without running additional, shorter simulations. The results of this project work support this claim in principle. The insensitivity does appear in the expectation-value comparison, where Alfonsi’s result drifts as the threshold is lowered while Duncan’s stays essentially fixed across its whole range. Both the difference in simulation duration and the relative insensitivity against the parameter selection can be explained by comparing the “stochastic share” of this two algorithms. Using the mean number of stochastic reaction events N as a measure for this share, the Duncan algorithm spends a consistently higher fraction of the simulation in the stochastic regime, which accounts for the runtime difference. On the other side, since the duncan algorithm spends a high fraction of its simulation time in the stochastic regime even for low parameter values it is reproducing the SSA results relatively independent of the exact parameter choice.

It is worth stressing that the two algorithms control fundamentally different quantities: for Alfonsi a single propensity threshold per reaction, for Duncan a molecule-number-based decision, via Eq (6), of whether a species counts as stochastic – essentially the product of per-species weights (each between 0 and 1) over every species involved in a reaction. At first sight this makes the comparison between the algorithms, questionable. What makes the comparison plausible again is that N follows the same increase over the respective parameter ranges for both schemes, so the ranges probed here do correspond to comparable operating points. Second, since the SSA result is in fact already reproduced at considerably smaller values the parameter range chosen include the sensible upper bounds for both the alfonsi as well as for the duncan parameters. Therefore even if a propensity threshold doesn’t translate directly to duncan threshold influence of the parameter values over the ranges of interest can be compared this way.

The claim of Duncan et al. that the Thinning variant permits larger step sizes could not be confirmed for this example: Duncan-SSA and Duncan-Thinning reproduce the SSA up to the same step size. One possible explanation for this (the original paper didn’t consider at all) is this that the step size is capped by the accuracy of the CLE itself – a limit that Thinning is not designed to relax – rather than by the thinning procedure. To confirm this the two step-size bounds were estimated separately. The bound set by CLE

stability turned out to be the *less* restrictive of the two, lying above the bound set by the propensity-constancy assumption (S1 Tab); the step size is therefore capped by propensity constancy. This therefore confirms the theoretical advantage of the Thinning method in this case and still does not explain why it shows no advantage over Duncan-SSA in the range below that bound, where both are valid.

The findings come with several limitations that should be briefly mentioned here. The propensity-constancy bound rests on an allowed propensity deviation of 5%, chosen fairly arbitrarily rather than derived from the target accuracy of a specific application; a different margin would probably shift the reported bound. The mean number of reaction events per simulation is likewise only a proxy (it is not trivial to define a better one since duncan has one stochastic and one hybrid regime, while alfonsi only divides into one stochastic one). Finally, the two algorithms were compared on only two test systems over specific parameter ranges, so the relative ranking found here are not proven to carry over to systems with different stoichiometry or kinetics.

Several questions raised by these results were beyond the scope of this project and represent potentially interesting future research topics. Until now it remains unexplained why the boundary-condition effect that dominates the Lotka–Volterra comparison disappears in the bacteriophage model, although that model too is simulated with a boundary at zero; understanding this would clarify how boundary handling actually matters for the CLE and what its principal limitations are. A second open problem lies in an adaptive step-size implementation of the duncan algorithm, which chooses Δt during the simulation from a local error estimate. The need of finding an appropriate step size before every simulation is a clear downside of the current Duncan Framework and an adaptive scheme could potentially resolve this need while improving run times simultaneously. Because of the independence of the constant propensity condition, a Duncan-Thinning implementation for the adaptive step size control could focus on local error considerations only and is again expected to be advantageous here. Another interesting question is whether the notion of a convergence order developed for classical SDE solvers with fixed step sizes [12] can be extended to this class of hybrid schemes and, if so, what order each of the algorithms compared here would attain. Additionally it would be informative to break the runtime difference between Alfonsi and Duncan down into its components – the cost of Duncan’s CLE step against that of Alfonsi’s deterministic rate-equation step, and the fixed per-step overhead each algorithm adds – to see exactly where the observed difference in time complexity comes from in addition to the differing stochastic shares of their simulation. Finally as stated before the propensity-constancy test would benefit from a more rigorous statistical treatment: the present version fixes the equivalence margin and the sample size by hand, and it is not clear how strongly the resulting step-size bound depends on either choice, so a systematic study of that dependence would make the bound more trustworthy.

Supporting information

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S1 Text. Validation of the hybrid implementations against analytically solvable models. Before either hybrid algorithm was applied to the two test models of this report, both implementations were checked against systems whose expectation values and probability distributions are known in closed form. The point of such a check is that the reference is exact: any discrepancy can be attributed to the implementation itself rather than to the model, to the ensemble size or to a competing approximation. Two systems were used, one with a single species and an unbounded state space, one with two species and a finite state space.

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Birth–death process. The first system consists of a species A produced at a constant rate and degraded in proportion to its abundance,

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Its mean obeys a linear ordinary differential equation and therefore relaxes exponentially from the initial value to the ratio of the two rate constants,

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$$\langle A(t) \rangle = \frac{k_1}{k_2} + \left(A(0) - \frac{k_1}{k_2} \right) e^{-k_2 t}. \quad (12)$$

The stationary distribution follows from the detailed-balance recursion of the master equation, $\pi_k k k_2 = \pi_{k-1} k_1$, which gives $\pi_k = \pi_0 \prod_{i=1}^k k_1 / (i k_2)$. Normalisation fixes π_0 to $\pi_0 = e^{-k_1/k_2}$ and hence the stationary distribution is a Poisson distribution with parameter $\frac{k_1}{k_2}$:

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$$\pi_k = \frac{(k_1/k_2)^k}{k!} e^{-k_1/k_2}. \quad (13)$$

Both quantities were estimated from 10^3 trajectories. The expectation value was evaluated on a time grid of width 0.05 over the whole horizon; the stationary distribution was obtained by pooling the states over $t \in [10, 20]$ on the same grid, that is, over a window in which the system has long since relaxed, so that the time average may stand in for the ensemble average of the stationary state.

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Reversible dimerisation. The second system couples two species through a bimolecular forward and a monomolecular backward reaction,

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Here the quantity $M = n_A + 2n_B$ is conserved, so the reachable states are the finitely many pairs $(M, 0), (M-2, 1), \dots, (0, \lfloor M/2 \rfloor)$ and can be indexed by $n_B = i$ alone. The chemical master equation thus reduces to a finite linear system $\dot{\mathbf{P}}(t) = \mathbf{A} \mathbf{P}(t)$ with:

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$$A_{i+1,i} = k_1 \frac{(M-2i)(M-2i-1)}{2}, \quad A_{i-1,i} = i k_2, \quad A_{i,i} = - \left(k_1 \frac{(M-2i)(M-2i-1)}{2} + i k_2 \right) \quad (15)$$

A finite linear system of this form is solved exactly by the eigendecomposition of $\mathbf{A}$, $\mathbf{P}(t) = \sum_i c_i \mathbf{v}_i e^{\lambda_i t}$, with the coefficients c_i determined by the initial distribution. The expectation value of A then follows from the conservation law,

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$$\langle A(t) \rangle = \sum_k (n_A(0) + 2(n_B(0) - k)) P_k(t), \quad (16)$$

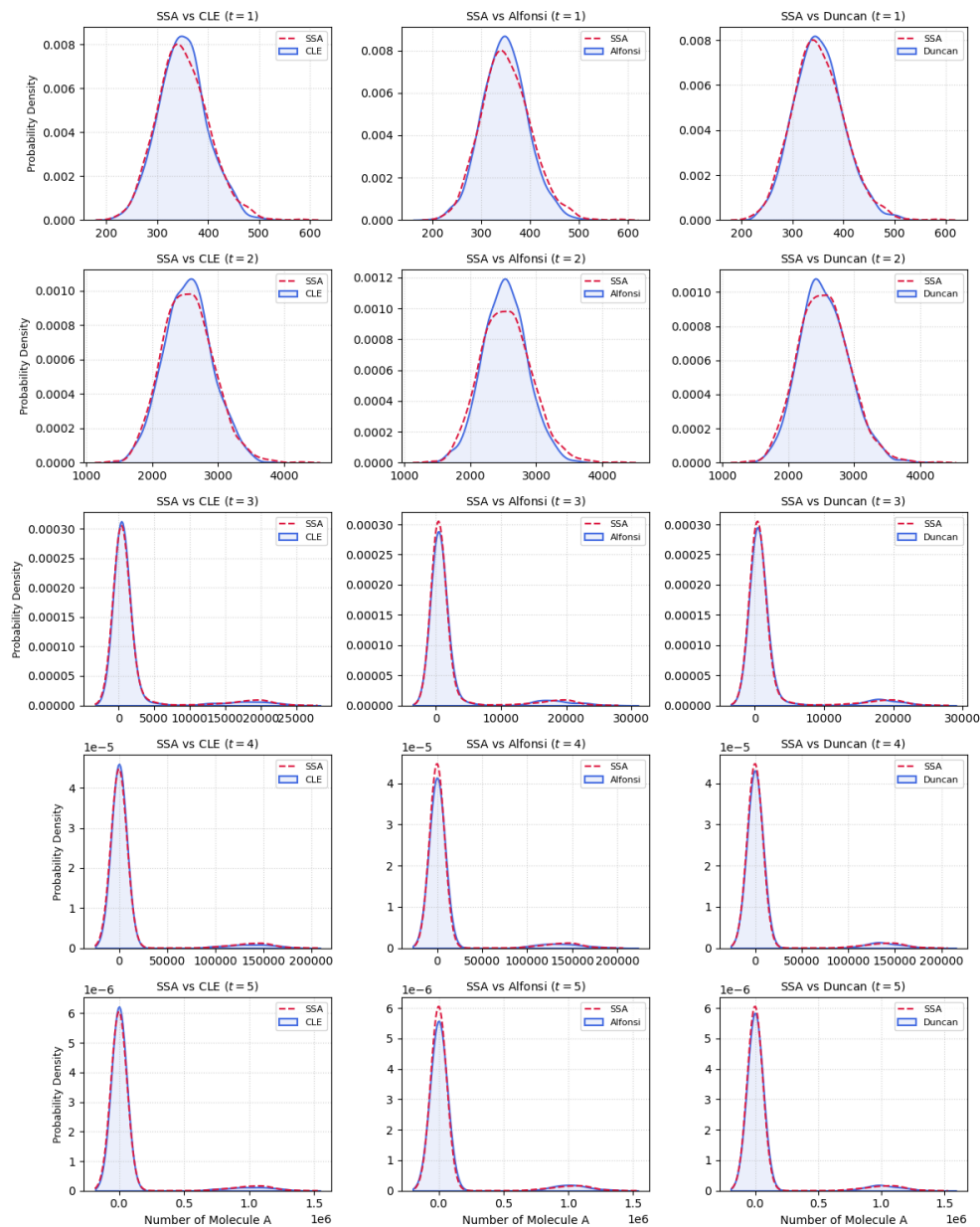
where $P_k(t)$ denotes the k -th component of $\mathbf{P}(t)$, that is, the probability of the state with $n_B = k$ and correspondingly $n_A = M - 2k$. The expectation value was compared with the average over 10^3 simulated trajectories on a grid of 10^3 points over $t \in [0, 2]$. One detail matters when the system size is varied: only the bimolecular constant k_1 scales inversely with the reaction volume, while k_2 does not, so a change of scale must be applied to k_1 alone if the ratio of the two volume-independent constants is to be preserved.

The settings of both checks are collected in Table 5. With them, the Alfonsi implementation and the Duncan-SSA implementation each reproduced the analytical mean of both systems and the Poisson stationary distribution of the birth–death process, and were used for the experiments of this report on that basis.

Table 5. Settings of the validation runs against the two analytically solvable models.

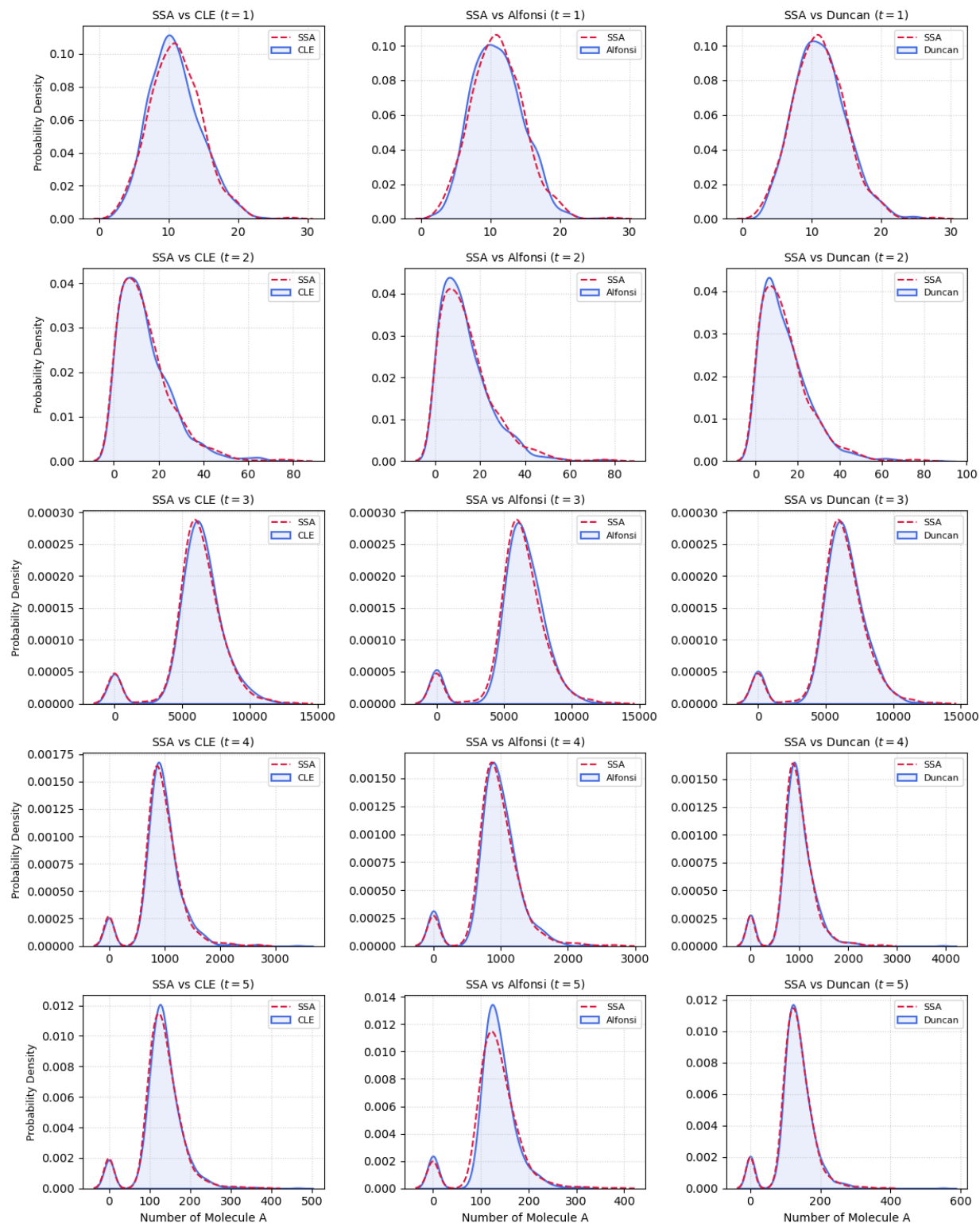
System	Setting	Value
Birth–death	k_1, k_2	2.0, 1.0
	$A(0)$, horizon	100, $t \in [0, 20]$
	ensemble, output grid	10^3 , 0.05
	α_{thresh} (Alfonsi)	10
	$\delta t, \Delta t, [I_1, I_2]$ (Duncan)	10^{-2} , 10^{-3} , 25/35
Dimerisation	k_1, k_2	0.1, 5.0
	$n_A(0), n_B(0)$, horizon	40, 40, $t \in [0, 2]$
	ensemble, output grid	10^3 , 10^3 points
	α_{thresh} (Alfonsi)	30
	$\delta t, \Delta t, [I_1, I_2]$ (Duncan)	10^{-2} , 10^{-3} , 25/35

S1 Fig. One-dimensional marginal probability distribution of molecule *A*, Lotka–Volterra model. Marginal of the joint distribution shown in Fig 4, under the absorbing boundary condition, at $t \in \{3, 4, 5\}$.



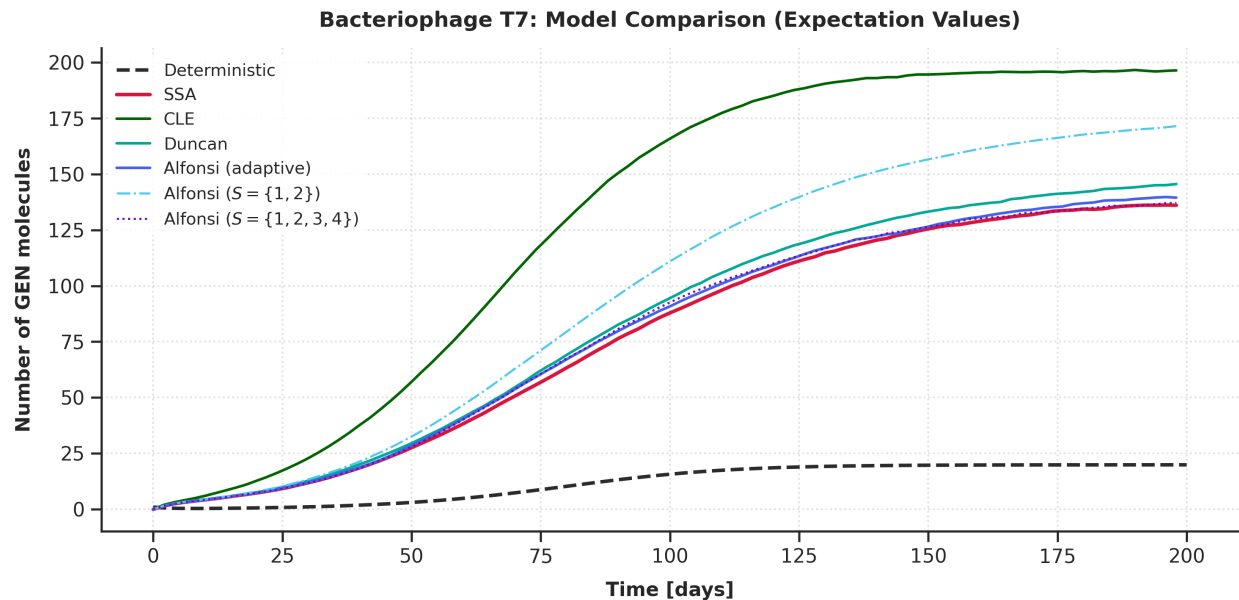
S2 Fig. One-dimensional marginal probability distribution of molecule B , Lotka–Volterra model. Marginal of the joint distribution shown in Fig 4, under the absorbing boundary condition, at $t \in \{3, 4, 5\}$.

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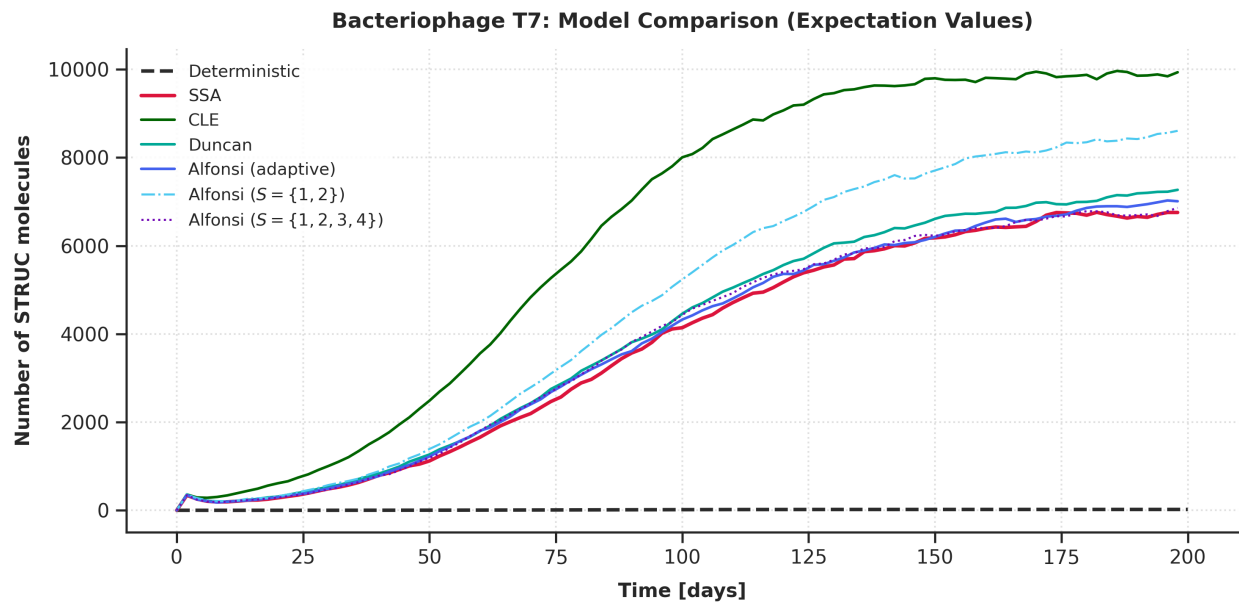
S3 Fig. Expectation value of the genomic species (gen), bacteriophage T7 model. SSA, Alfonsi, Duncan-SSA and the CLE, complementing the template-species comparison in Fig 5.

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S4 Fig. Expectation value of the structural species (struc), bacteriophage T7 model. SSA, Alfonsi, Duncan-SSA and the CLE, complementing the template-species comparison in Fig 5.

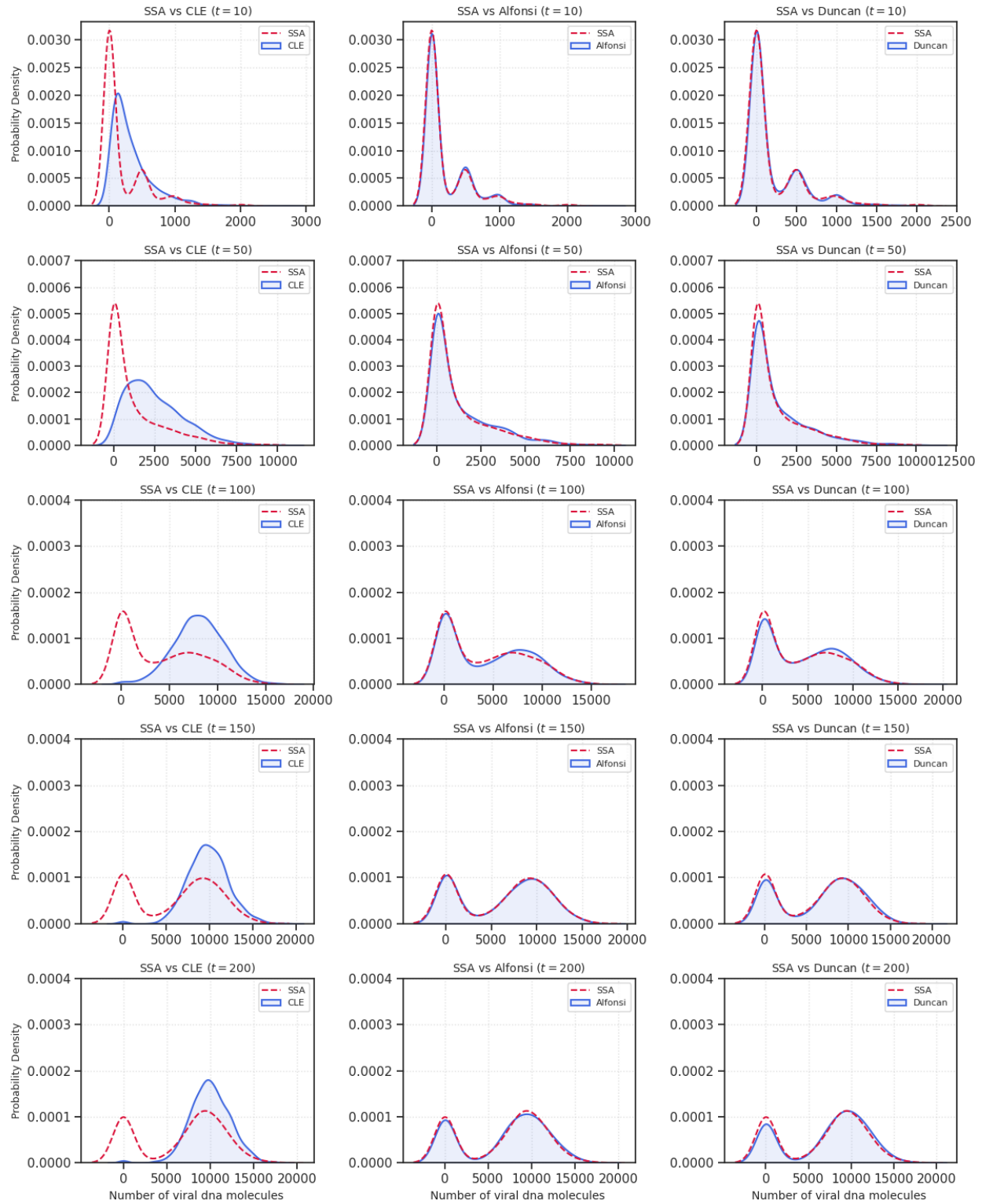
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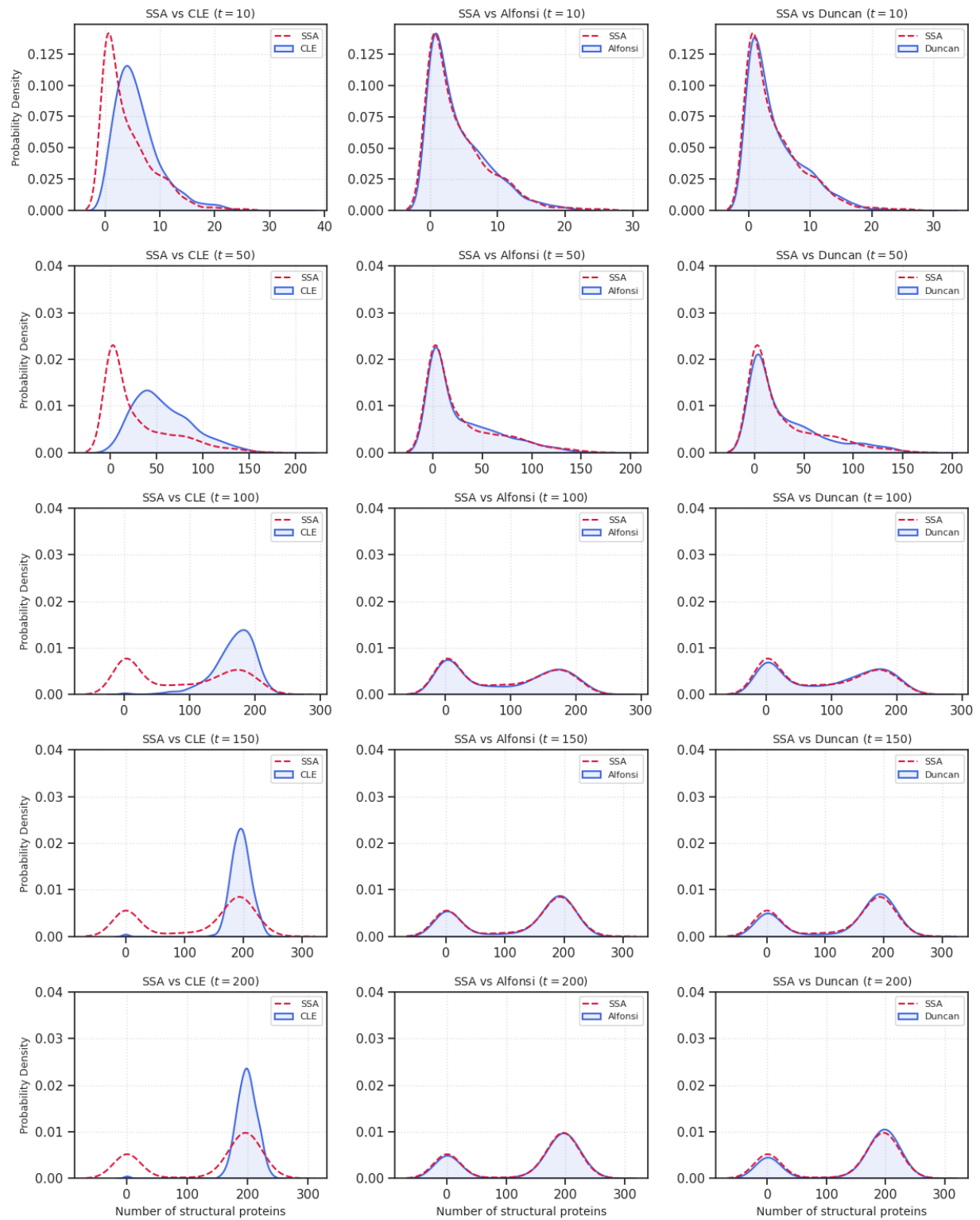
S5 Fig. Probability distribution of the genomic species (gen), bacteriophage T7 model. SSA, Alfonsi, Duncan-SSA and the CLE, complementing the template-species comparison in Fig 6.

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S6 Fig. Probability distribution of the structural species (struc), bacteriophage T7 model. SSA, Alfonsi, Duncan-SSA and the CLE, complementing the template-species comparison in Fig 6.



S1 File. Simulation source code. Python implementation of the SSA, the CLE and the three hybrid algorithms (Alfonsi, Duncan-SSA, Duncan-Thinning) in `stochastinetics.py`, together with all the scripts that generate the figures and tables of this report. Available at <https://github.com/JonaSimonScheibe/hybrid-ssa-comparison/tree/main>.

S1 Tab. Results of the TOST equivalence test for different time steps Δt and reactions.

Δt	Reaktion	Mean rel. deviation	$p_{\text{lower}} / p_{\text{upper}}$	Equivalence ($\pm 5\%$)
0.001	0	−0.0023	0.0 / 0.0	True
	1	−0.0003	0.0 / 0.0	True
	2	0.0019	0.0 / 0.0	True
0.00316	0	−0.0070	0.0 / 0.0	True
	1	−0.0008	0.0 / 0.0	True
	2	0.0062	0.0 / 0.0	True
0.01	0	−0.0212	0.0 / 0.0	True
	1	−0.0007	0.0 / 0.0	True
	2	0.0207	0.0 / 0.0	True
0.0316	0	−0.0584	1.0 / 0.0	False
	1	0.0163	0.0 / 0.0	True
	2	0.0752	0.0 / 1.0	False
0.1	0	−0.1185	1.0 / 0.0	False

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